LESSON 102 (1.5 GROUND)

Lesson Objective

During this lesson, the student will learn how to obtain a weather briefing and complete a takeoff and landing data (TOLD) card. This lesson is intended to give the student a thorough understanding of the requirements for preflight planning.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
    - Ask the student what they learned from the Pilot's Handbook of Aeronautical Knowledge chapter 10 - Weight and Balance.
    - Review the FOM quiz.
- Student questions.

References

- Advisory Circulars. (Search Tool)
    - Aviation Weather. (AC 00-6B)
    - Aviation Weather Services. (AC 00-45H)
- Pilot's Handbook of Aeronautical Knowledge. (FAA-H-8083-25B)
    - Chapter 10 - Weight and Balance.
    - Chapter 11 - Aircraft Performance.
    - Chapter 12 - Weather Theory.
    - Chapter 13 - Aviation Weather Services.
- Warrior III Pilot's Information Manual.
    - Section 05 - Performance.
    - Section 06 - Weight and Balance.

Lesson Content

METARs and TAFS

- Show the student how to decode METARs and TAFs and put the information on to the TOLD card.
    - Review issuance times and valid times.
    - Review what is included in the reports.

Obtaining a Weather Briefing

- Stress the importance of obtaining weather information prior to a flight (go/no-go decision).
- It is recommended to guide the student through a telephone/online briefing.
- Eventually a self-brief must progress beyond "weight and balance, METAR, TOLD card, and look outside".
    - Types of weather briefings:

        - Outlook.
        - Standard.
        - Abbreviated.
        - In-flight.
    - For the purpose of this lesson, show the student how to obtain a weather briefing by calling the briefer or online.
        - Telephone:
            - Call 1-800-992-7433 → say "briefer" → say "Southern Florida".
            - Give the briefer the flight plan information (use the reference on page 4 if needed) and request a standard weather briefing.
            - Have the weather products opened on aviationweather.gov during the briefing to aid student understanding.
        - Online:
            - Navigate to 1800wxbrief.com and have the student create an account.
            - Proceed to the plan & brief tab and input all of the required flight plan information (use the reference on page 4 if needed).
            - Click on route brief and select standard briefing.

Pilot Information Manual (PIM)

- Outline the different sections of the Pilot Information Manual.
- Discuss what can be found in each of the sections (GLENPWAAS).
    - **G**eneral.
    - **L**imitations.
    - **E**mergency procedures.
    - **N**ormal procedures.
    - **P**erformance.
    - **W**eight and balance.
    - **A**ircraft systems.
    - **A**ircraft servicing and handling.
    - **S**upplements.

Performance & Weight and Balance (TOLD card)

- TOLD Card:
    - Fill out all fields of a TOLD card with the student.
    - Explain the purpose of the TOLD cards:
        - Safety.
        - Preflight action (91.103).
- Weight and Balance (PIM Ch 06, PHAK Ch 10):
    - Introduce how to find arms and what they mean.
    - Show the student how to find CG and chart it on the envelope for takeoff and landing.

- Effects of CG on performance:
    - Forward CG:
        - Decrease in cruise speed.
        - Increase in stall speed.
        - Increase in longitudinal stability.
        - Easier to recover from a stall.
    - Aft CG:
        - Increase in cruise speed.
        - Decrease in stall speed.
        - Decrease in longitudinal stability.
        - Harder to recover from a stall.
- Effect of weight on performance:
    - Increase in weight:
        - Longer takeoff and landing distances.
        - Higher takeoff speeds.
        - Reduced rate and angle of climb.
        - Lower maximum altitude.
        - Shorter range.
        - Reduced cruising speed.
        - Higher stalling speed.
        - Higher approach and landing speed.
        - Excessive weight on nose/tailwheel.
- Discuss ballast if necessary.
- Show the student how to find taxi fuel burn in the PIM.

- Performance – (PIM Ch 05):
    - Show the student how to find all the V-Speeds in the PIM.
    - Discuss the atmospheric conditions' impact on performance.
        - Density and pressure altitude.
    - Introduce the crosswind component chart to the student.
    - Introduce the takeoff and landing charts (roll and 50-foot obstacle).

Post-Brief

- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 103.
- Assign homework:
    - Assign weight and balance practice problems.

Completion Standards (continued on next page)

This lesson is complete when the student is able to:

1. Obtain a weather briefing by utilizing 1-800-WX-BRIEF and complete a TOLD card with instructor guidance.
2. Understand aircraft performance charts with instructor guidance.
3. Calculate density altitude and apply it to performance calculations with instructor guidance.
4. Understand all environmental factors that affect performance.

References

3 MESSAGE TYPE — << ≡ (FPL
7 AIRCRAFT IDENTIFICATION — -
8 FLIGHT RULES — -
TYPE OF FLIGHT — << ≡

9 NUMBER — -
TYPE OF AIRCRAFT
WAKE TURBULENCE CAT. — /
10 EQUIPMENT — - / << ≡

13 DEPARTURE AERODROME — -
TIME — << ≡

15 CRUISING SPEED — - N 0
LEVEL
ROUTE

<< ≡

16 DESTINATION AERODROME — -
TOTAL EET HR. MIN
ALTN AERODROME
2ND ALTN AERODROME — << ≡

18 OTHER INFORMATION

) << ≡

SUPPLEMENTARY INFORMATION (NOT TO BE TRANSMITTED IN FPL MESSAGES)

19 ENDURANCE HR MIN — - E /
PERSONS ON BOARD — P /
EMERGENCY RADIO — R / UHF U, VHF V, ELT E

SURVIVAL EQUIPMENT — S / POLAR P, DESERT D, MARITIME M, JUNGLE J
JACKETS — J / LIGHT L, FLUORES F, UHF U, VHF V

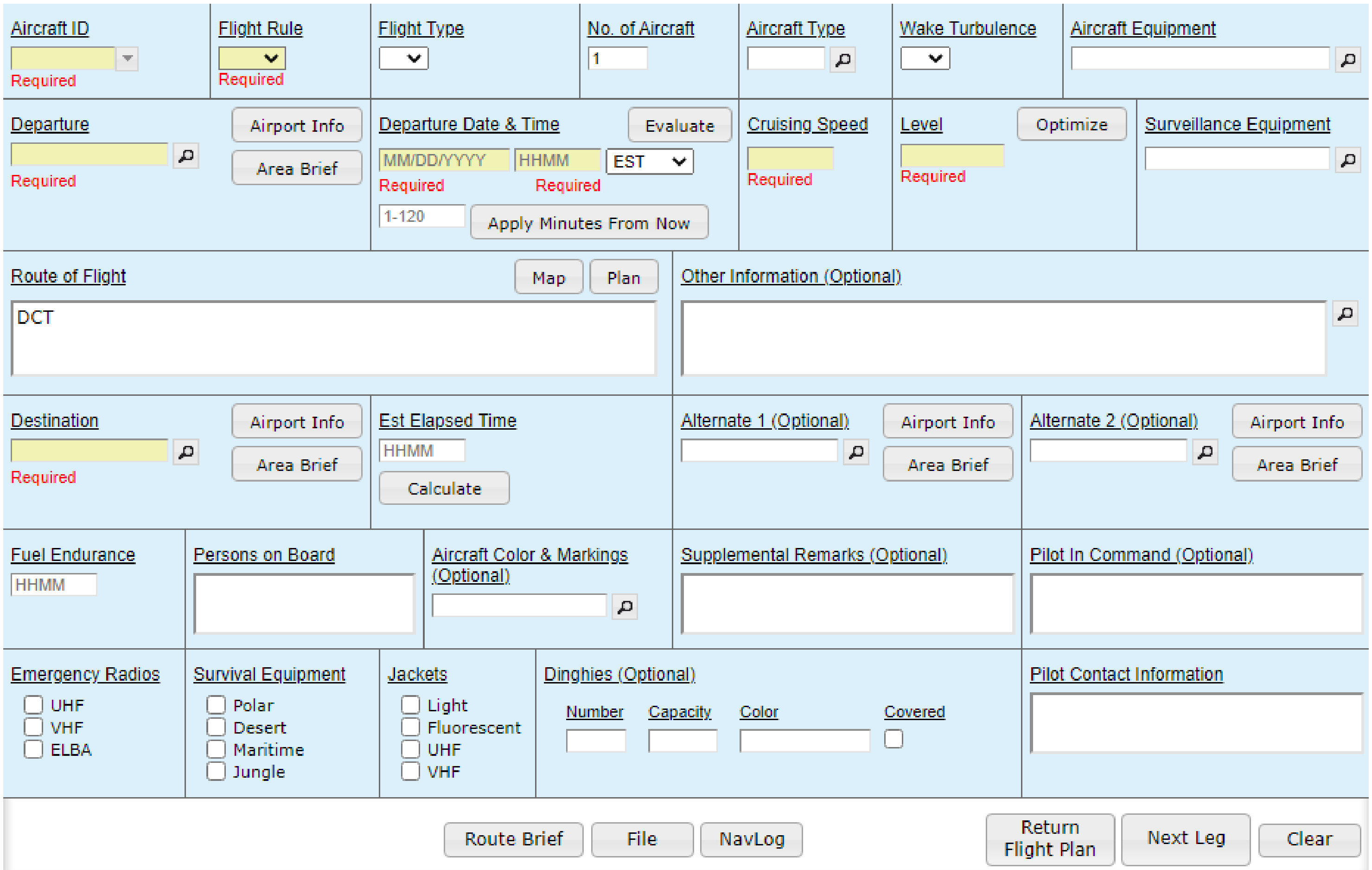
DINGHIES
D / NUMBER, CAPACITY, COVER C, COLOUR << ≡

AIRCRAFT COLOUR AND MARKINGS — A /

REMARK — N / << ≡

PILOT IN COMMAND — C /) << ≡

FILED BY
SPACE RESERVED FOR ADDITIONAL REQUIREMENTS



Aircraft ID
Required
Flight Rule
Required
Flight Type
No. of Aircraft
1
Aircraft Type
Wake Turbulence
Aircraft Equipment
Departure
Required
Airport Info
Area Brief
Departure Date & Time
Evaluate
MM/DD/YYYY
HHMM
EST
Required
Required
1-120
Apply Minutes From Now
Cruising Speed
Required
Level
Optimize
Required
Surveillance Equipment
Route of Flight
Map
Plan
DCT
Other Information (Optional)
Destination
Required
Airport Info
Area Brief
Est Elapsed Time
HHMM
Calculate
Alternate 1 (Optional)
Airport Info
Area Brief
Alternate 2 (Optional)
Airport Info
Area Brief
Fuel Endurance
HHMM
Persons on Board
Aircraft Color & Markings (Optional)
Supplemental Remarks (Optional)
Pilot In Command (Optional)
Emergency Radios
UHF
VHF
ELBA
Survival Equipment
Polar
Desert
Maritime
Jungle
Jackets
Light
Fluorescent
UHF
VHF
Dinghies (Optional)
Number
Capacity
Color
Covered
Pilot Contact Information
Route Brief
File
NavLog
Return Flight Plan
Next Leg
Clear